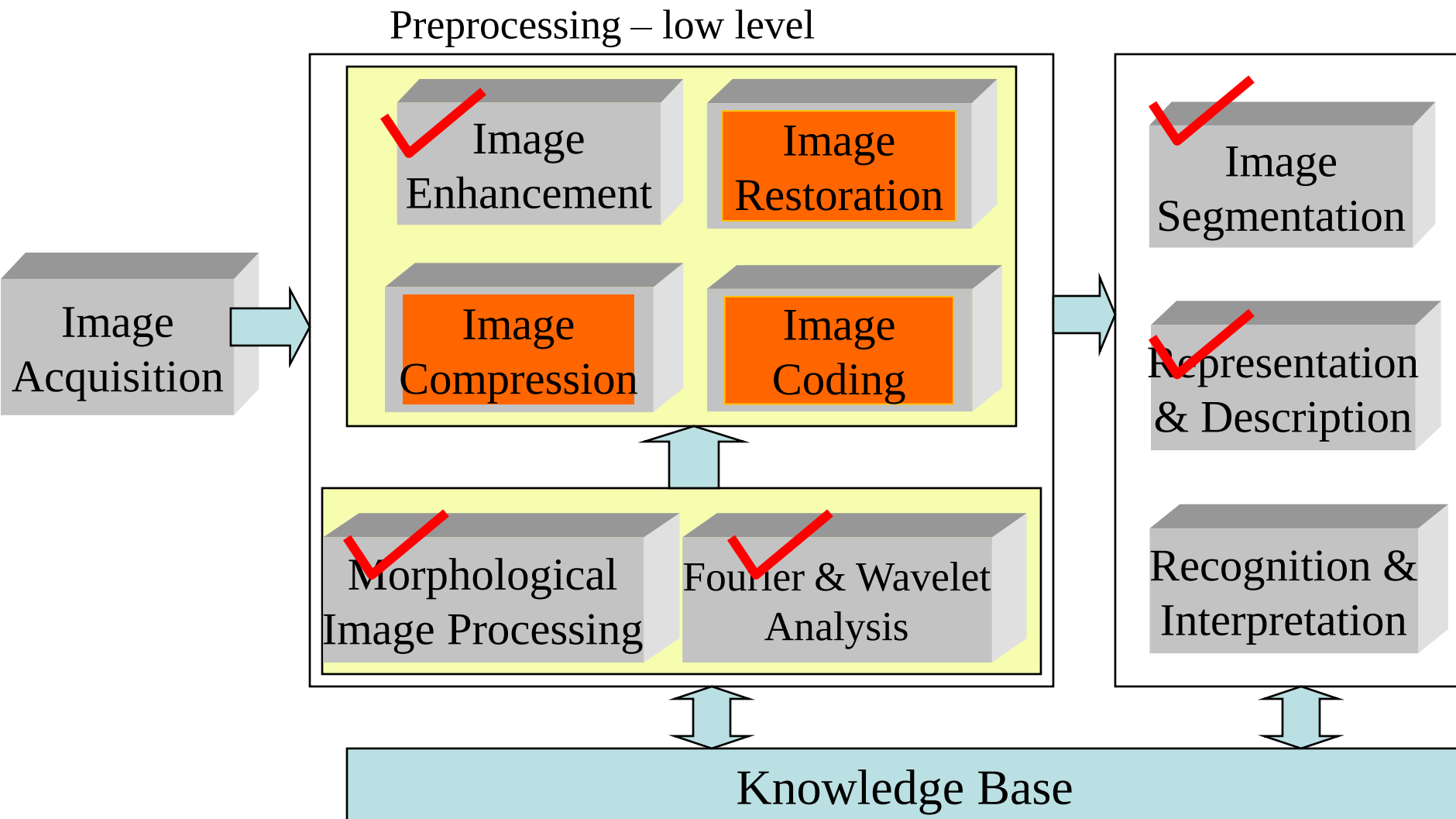


Digital Image Processing 10:

Representation and Description

CHENGLU WEN
Department of AI
clwen@xmu.edu.cn

DIP Components

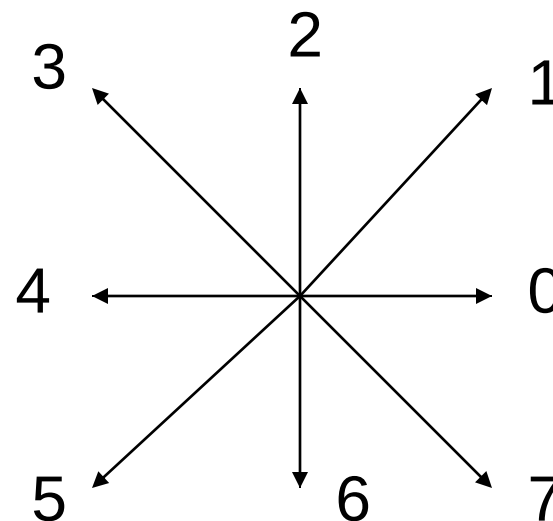
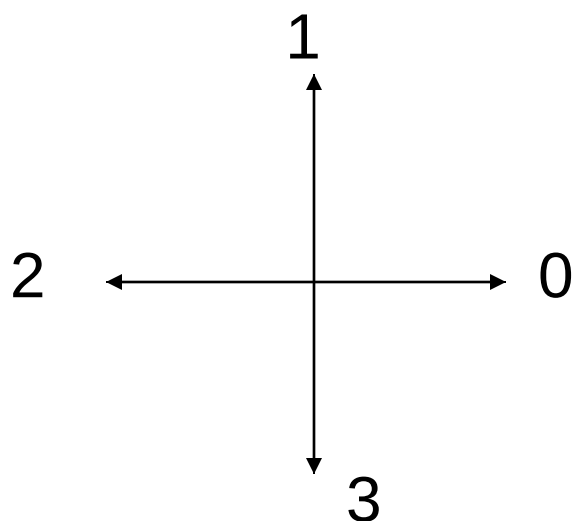


Tasks in image representation/description

- Boundary descriptors
 - Chain codes
 - Signatures
 - Length, diameter, shape numbers, Fourier descriptors
 - Moments of signature
- Region descriptors
 - Skeleton
 - Area, compactness, principal axes
 - Texture
 - Moment

Boundary- Chain codes

- Represent a boundary by a connected sequence of straight-line segments of specified length and direction
- 4-directional chain code
- 8-directional chain code



4-directional chain code:

8-directional chain code:

1200107666654444322

Chain code normalization

- Normalize for rotation – **first difference**
 - Counting (counterclockwise) the number of directions that separate two adjacent elements of the code
- Normalize for starting position – **shape numbers**
 - The first difference of smallest magnitude

4-directional chain code:

011000103033332322221211

First difference:

310300133130003130003130

Shape number:

000313000313031030013313

Boundary- Signatures

- A 1-D functional representation of a boundary
- Basic idea: reduce the boundary representation to a **1-D function**, which might be easier to describe than a 2-D boundary
- Approach: Use the **distance from the centroid to the boundary as a function of angle**

Example

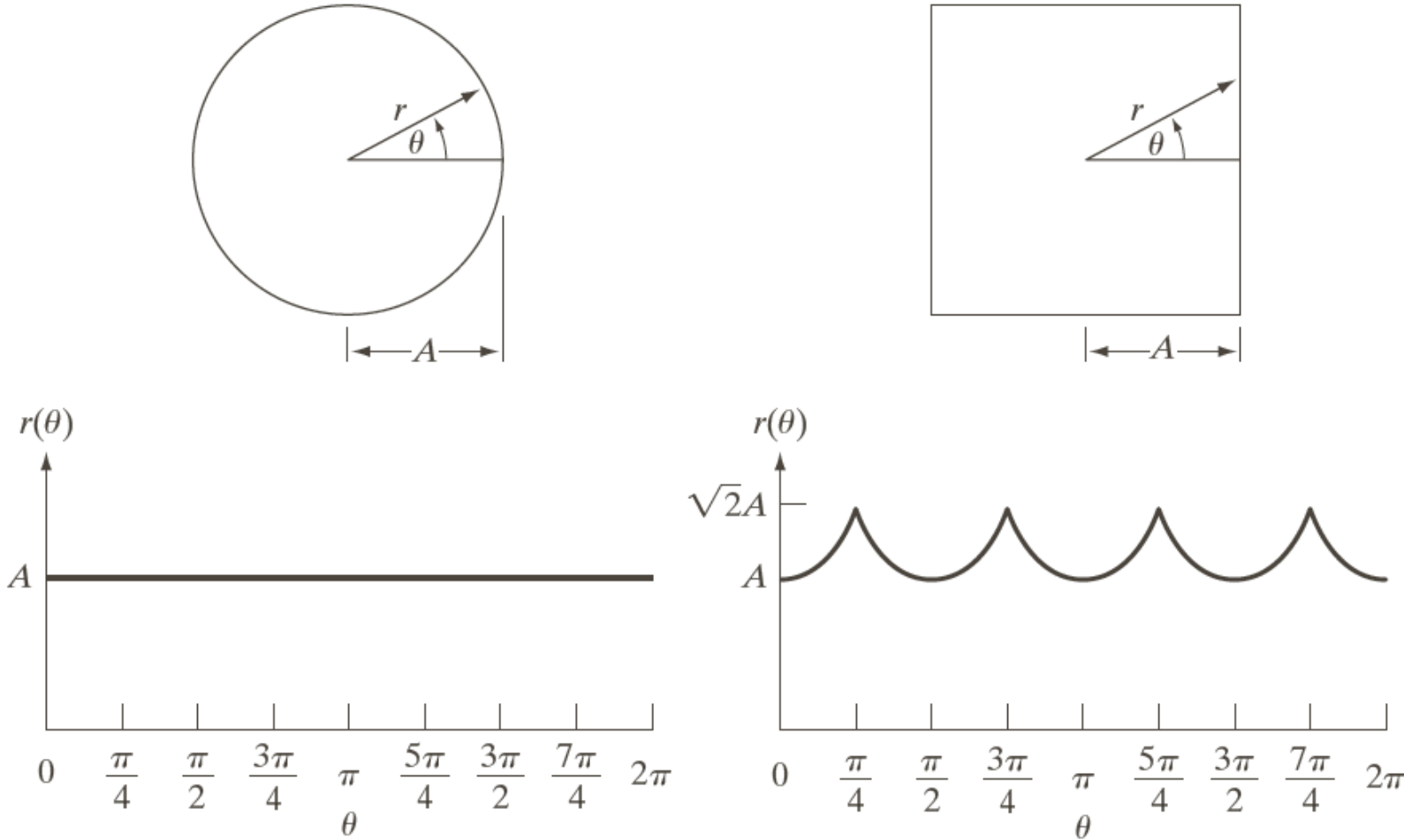
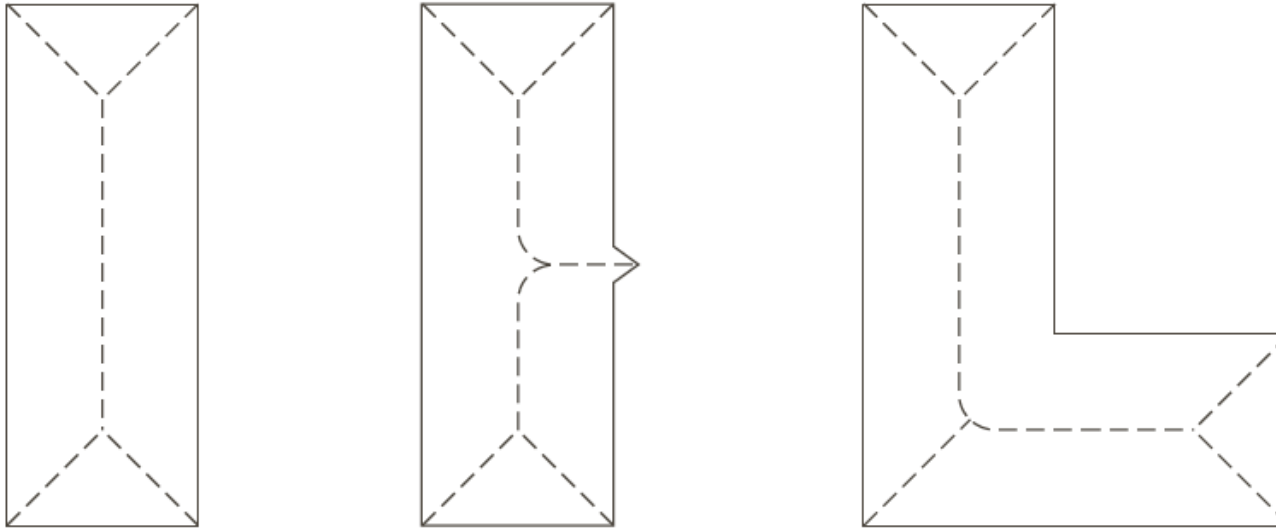


FIGURE 11.10
Distance-versus-angle signatures. In (a) $r(\theta)$ is constant. In (b), the signature consists of repetitions of the pattern $r(\theta) = A \sec \theta$ for $0 \leq \theta \leq \pi/4$ and $r(\theta) = A \csc \theta$ for $\pi/4 < \theta \leq \pi/2$.

Boundary- Skeleton of a region

- Use skeleton to represent a region
- Skeletonizing (thinning) algorithm
 - medial axis transformation (MAT): for each point (p) in region R, find its closest neighbor in the boundary (B). If it has more than 1 closest neighbor in B, then this point belongs to the media axis of the region (skeleton)
 - Computationally expensive

Example



a b c

FIGURE 11.13
Medial axes
(dashed) of three
simple regions.

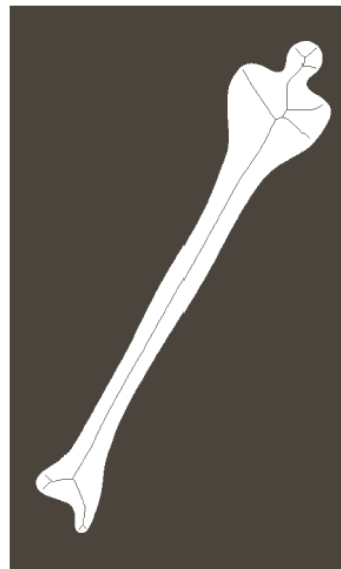


FIGURE 11.16
Human leg bone
and skeleton of
the region shown
superimposed.

Boundary- Simple boundary descriptors

- Length
 - for a chain-coded curve with unit spacing
 - length = **the number of vertical and horizontal components + $2^{0.5}$ times the number of diagonal components**
- Diameter
 - Maximum distance between any two points on the boundary
 - The line formed by these two points is called the **major axis of the boundary**
- Shape numbers

Boundary- Fourier descriptors

- Points on the boundary can be treated as a complex number:
- $s(k) = x(k) + jy(k)$
- Fourier descriptor: the discrete Fourier transform (DFT) of $s(k)$:

$$a(u) = \frac{1}{N} \sum_{k=0}^{N-1} s(k) \exp[-j2\pi uk / N]$$

- Usually, only the first few coefficients are used to represent the shape ($k=0:M$, $M < N$)

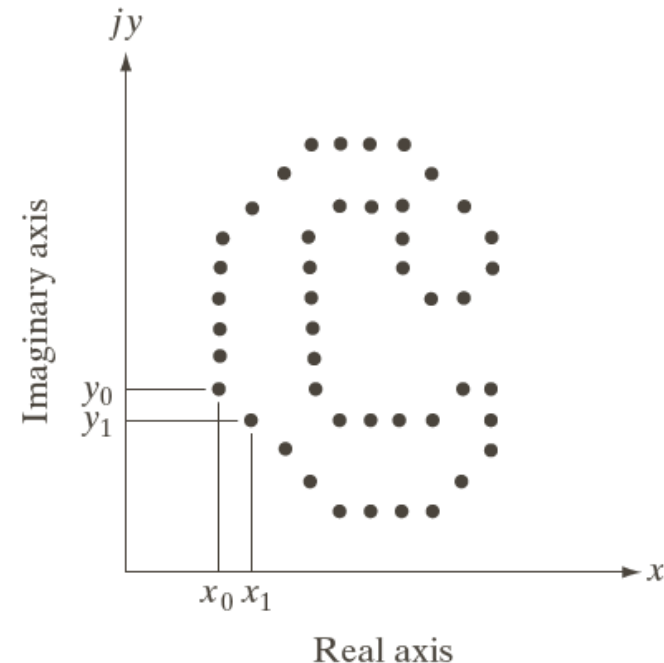
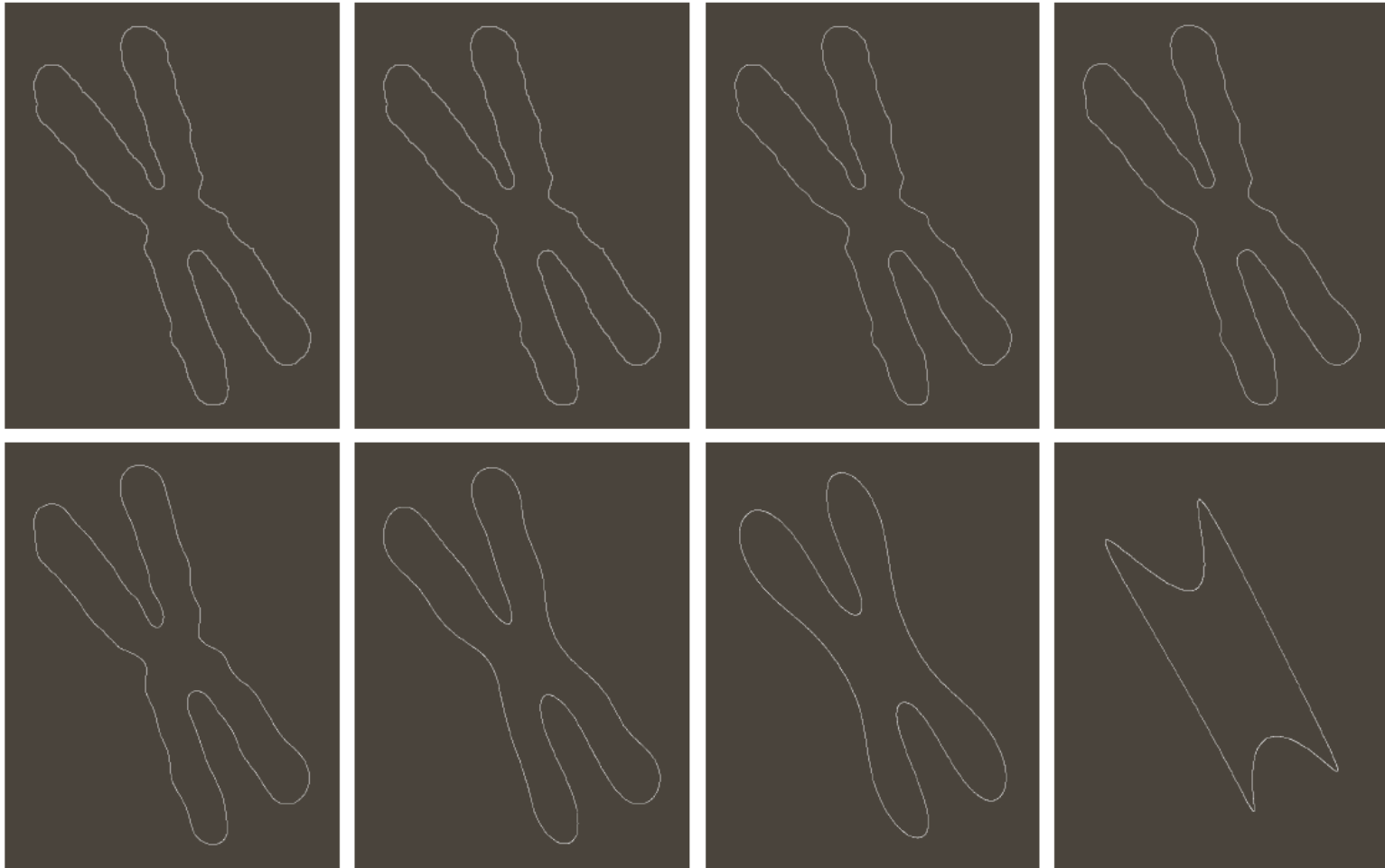


FIGURE 11.19

A digital boundary and its representation as a complex sequence. The points (x_0, y_0) and (x_1, y_1) shown are (arbitrarily) the first two points in the sequence.

Example



a	b	c	d
e	f	g	h

FIGURE 11.20 (a) Boundary of human chromosome (2868 points). (b)–(h) Boundaries reconstructed using 1434, 286, 144, 72, 36, 18, and 8 Fourier descriptors, respectively. These numbers are approximately 50%, 10%, 5%, 2.5%, 1.25%, 0.63%, and 0.28% of 2868, respectively.

Region-simple region descriptors

- Area
 - The number of pixels contained within its boundary
- Compactness
 - $\text{perimeter} * \text{perimeter} / \text{area}$
 - Insensitive to scale changes
 - Insensitive to orientation

Region-Texture descriptor

➤ Statistical approach

- Describe the smoothness, coarseness, and graininess of a region

➤ Structural approach

- Describe the arrangement

➤ Spectral approach

- Use Fourier spectrum to detect global periodicity

Texture -Statistical approach

- Use moments of the gray-level histogram to describe the texture

$$\mu_n(z) = \sum_{i=1}^L (z_i - m)^n p(z_i)$$

$p(z_i)$: probability

$$m = \sum_{i=1}^L z_i p(z_i): \text{ mean}$$

Texture -Special moments

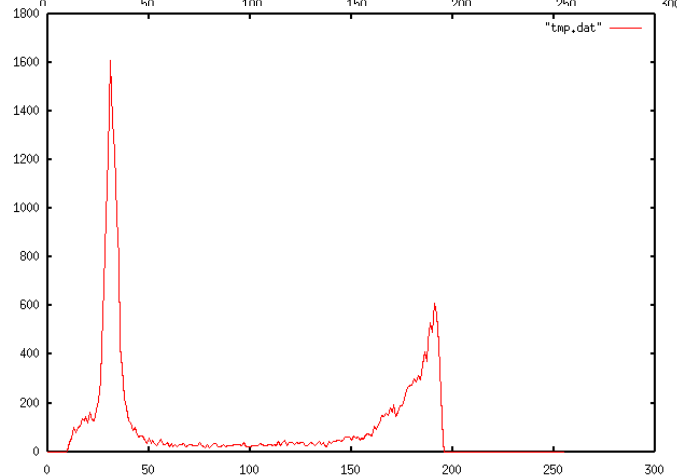
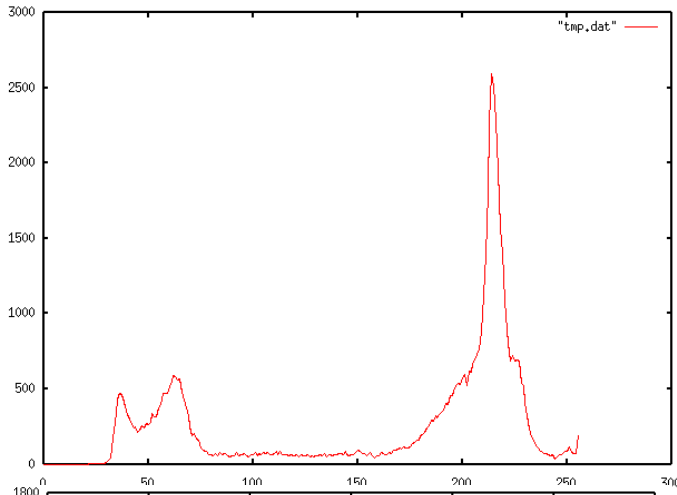
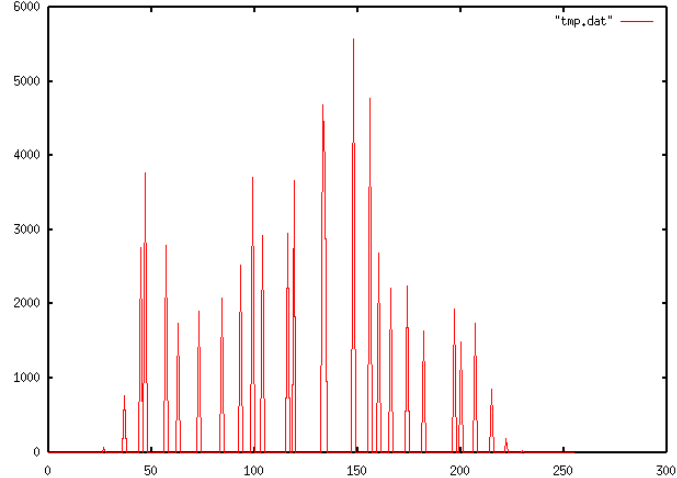
□ $\mu_0 = 1$

□ $\mu_1 = 0$

- The second moment: $[\sigma^2(z)]$ is the variance
 - Can be used to describe smoothness

$$R = 1 - \frac{1}{1 + \sigma^2(z)}$$

- The third moment (skewness): measure **skewness** of histogram
- The fourth moment (kurtosis): measure the **flatness** of histogram



lena.pgm

0	1
1	-1.05193e-06
2	2238.26
3	-9320.46
4	1.0756e+07

cameraman.pgm

0	1
1	7.68155e-06
2	4702.99
3	-281886
4	4.56105e+07

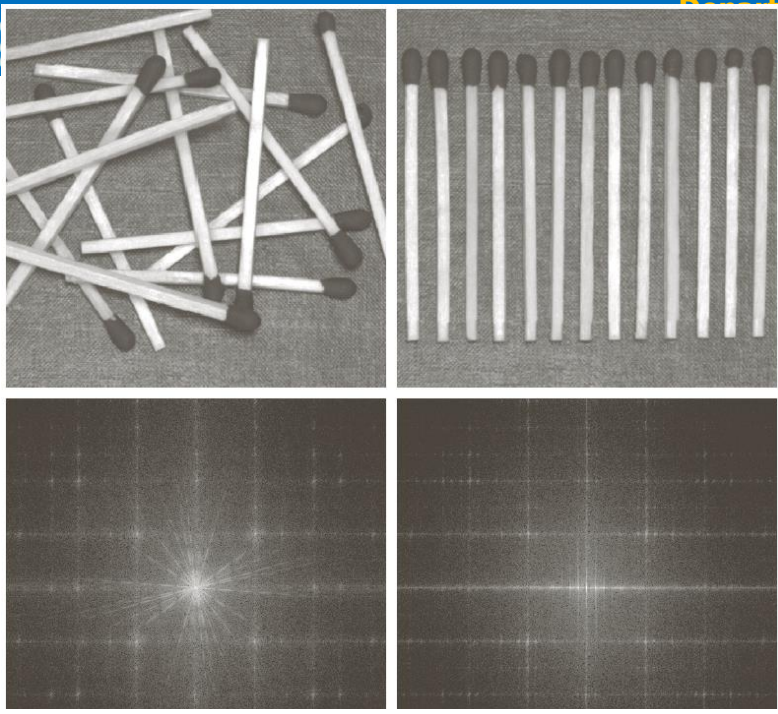
chess.pgm

0	1
1	-6.93423e-06
2	4988.22
3	101592
4	3.07141e+07

Region-Spectral approach

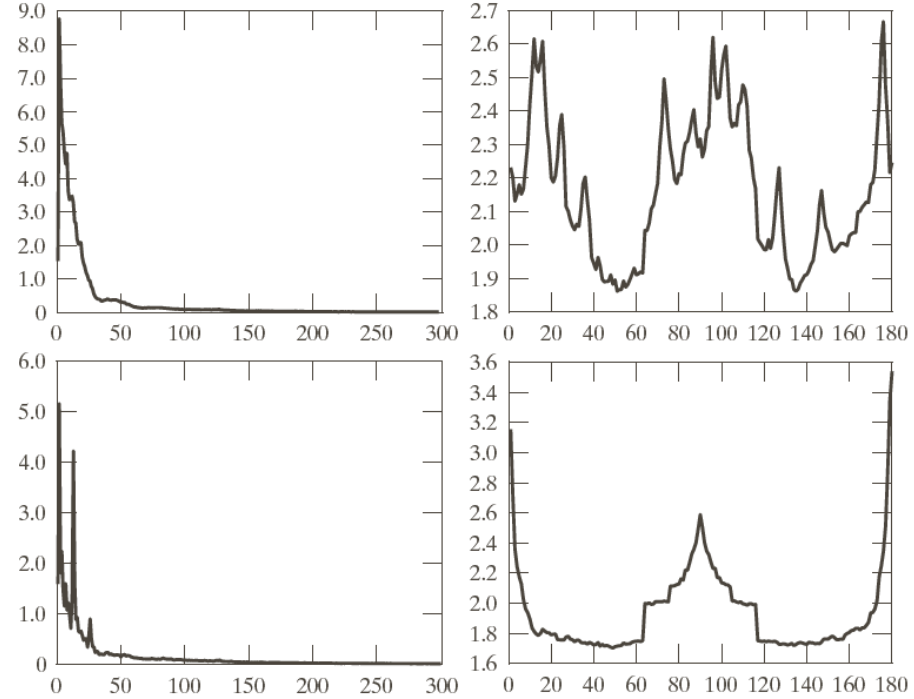
- Fourier spectrum is ideally suited for describing the directionality of periodic or almost periodic **2-D patterns** in an image
- Global texture patterns can be easily distinguishable as concentrations of high-energy bursts in the spectrum
 - **Prominent peaks** give the **principal direction** of the texture patterns
 - The **location of the peaks** gives the **fundamental spatial period** of the patterns
 - **Eliminating any periodic components** via filtering leaves nonperiodic image elements which can then be described by statistical techniques

$$S(r) = \sum_{\theta=0}^{\pi} S_{\theta}(r)$$
$$S(\theta) = \sum_{r=1}^R S_r(\theta)$$



a	b
c	d

FIGURE 11.35
(a) and (b) Images of random and ordered objects. (c) and (d) Corresponding Fourier spectra. All images are of size 600 × 600 pixels.



a	b
c	d

FIGURE 11.36
Plots of (a) $S(r)$ and (b) $S(\theta)$ for Fig. 11.35(a). (c) and (d) are plots of $S(r)$ and $S(\theta)$ for Fig. 11.35(b). All vertical axes are $\times 10^5$.

Region-Moments

- Moment sequence (m_{pq}):

$$m_{pq} = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} x^p y^q f(x, y)$$

- Uniqueness theorem: if $f(x,y)$ is piecewise continuous and has nonzero values only in a finite part of the xy plane, moments of all orders exist and the **moment sequence (m_{pq}) is uniquely determined by $f(x,y)$** . Conversely, (m_{pq}) uniquely determines $f(x,y)$
- Central moments:

$$\mu_{pq} = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} (x - \bar{x})^p (y - \bar{y})^q f(x, y), \quad \bar{x} = \frac{m_{10}}{m_{00}}, \quad \bar{y} = \frac{m_{01}}{m_{00}}$$

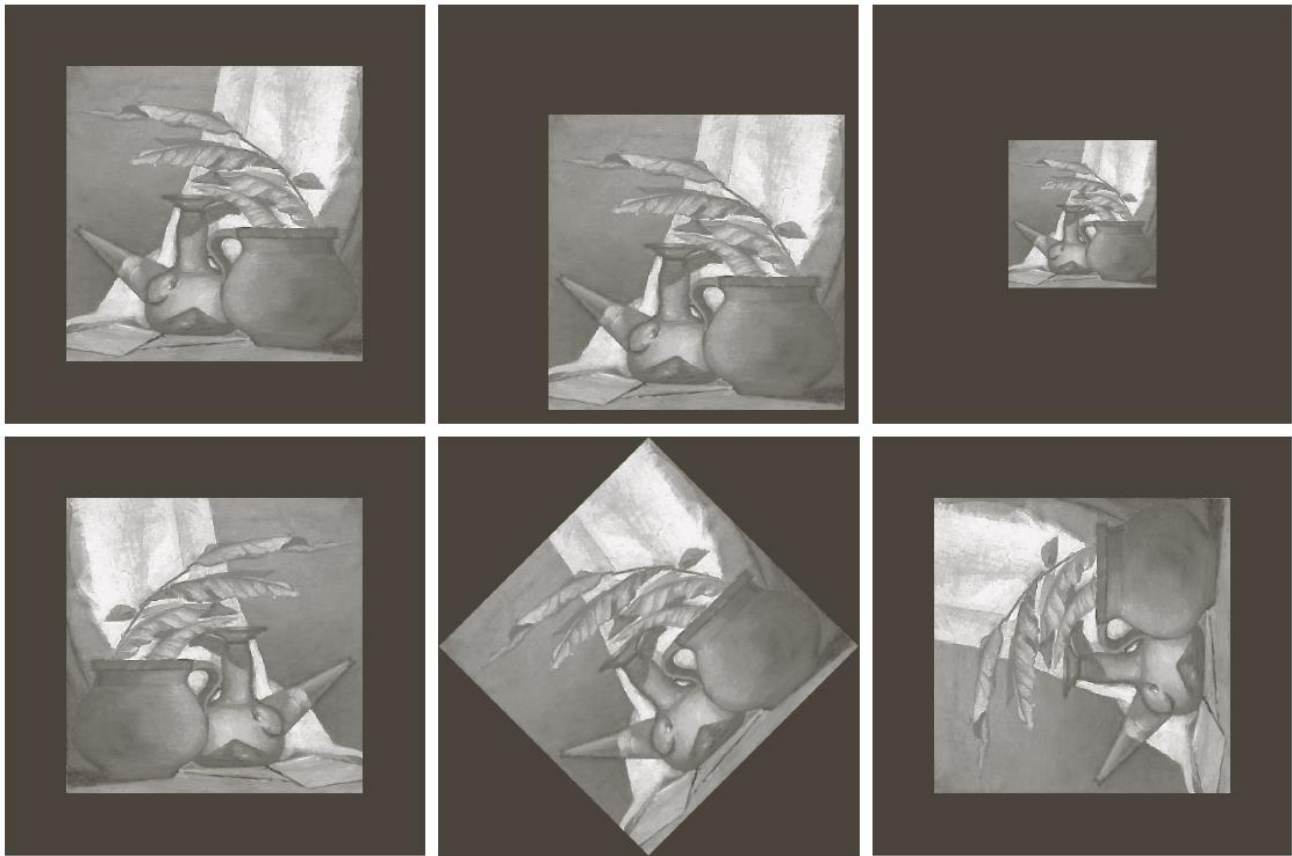
Invariant moments

- Invariant to translation, rotation, and scale change (η is the normalized central moment)

$$\phi_1 = \eta_{20} + \eta_{02}$$

$$\phi_2 = (\eta_{20} - \eta_{02})^2 + 4\eta_{11}^2$$

$$\vdots$$



a b c
d e f

FIGURE 11.37 (a) Original image. (b)–(f) Images translated, scaled by one-half, mirrored, rotated by 45° and rotated by 90°, respectively.

Moment Invariant	Original Image	Translated	Half Size	Mirrored	Rotated 45°	Rotated 90°
ϕ_1	2.8662	2.8662	2.8664	2.8662	2.8661	2.8662
ϕ_2	7.1265	7.1265	7.1257	7.1265	7.1266	7.1265
ϕ_3	10.4109	10.4109	10.4047	10.4109	10.4115	10.4109
ϕ_4	10.3742	10.3742	10.3719	10.3742	10.3742	10.3742
ϕ_5	21.3674	21.3674	21.3924	21.3674	21.3663	21.3674
ϕ_6	13.9417	13.9417	13.9383	13.9417	13.9417	13.9417
ϕ_7	−20.7809	−20.7809	−20.7724	20.7809	−20.7813	−20.7809

TABLE 11.5
Moment invariants for the images in Fig. 11.37.